

Jiali Shan

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Education

B.S. **Fudan University**, Mathematics and Applied Mathematics 2021 – 2025

Experience

University of California, Berkeley, Visiting scholar Berkeley, CA, USA
Aug 2024 – Dec 2024

Talks

Backward analysis of the total least squares and its Randomized Algorithms

- the 8th Peking University Symposium for Outstanding PhD Students in Computational and Applied Mathematics, Peking University, Nov. 2025.
- the 7th LuoJia Doctoral Forum in Mathematics, Wuhan University, Apr. 2026.

Funding

National Natural Science Foundation of China (NSFC)

- Category: Youth Student Basic Research Project
- Grant: 123B1008
- duration: 2023-2025
- topic: Randomized Algorithms for Toeplitz Total Least Squares Problems

Awards

2025 ICCM Undergraduate Paper Award Jan 2026

Skills

Coding: Matlab, Python, C

Languages: Mandarin Chinese (native), English (fluent), Japanese (basic)

Publications

Optimal backward error of a total least squares and its randomized algorithms Sept 2025

Jiali Shan, Yimin Wei

[10.1137/25M1749657](#) (SIAM Journal on Matrix Analysis and Applications)

Efficient estimate for the optimal backward error of the multidimensional total least squares 2026

Jiali Shan, Yimin Wei

(SIAM Journal on Matrix Analysis and Applications, to appear)

Neural networks for total least squares solution of the time-varying linear systems Sept 2025

Xuezhong Wang, Jiali Shan, Yimin Wei

[10.1007/s40314-024-03070-1](#) (Computational and Applied Mathematics)